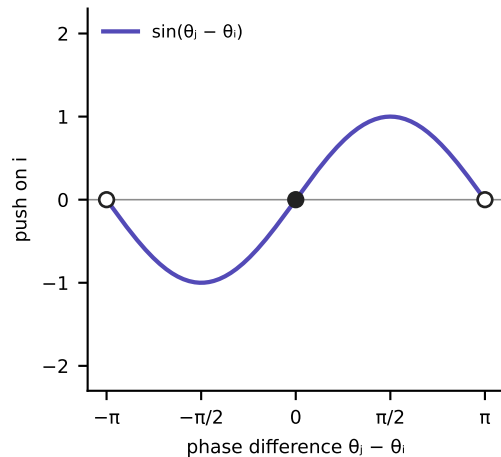
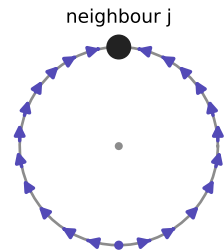


Kuramoto

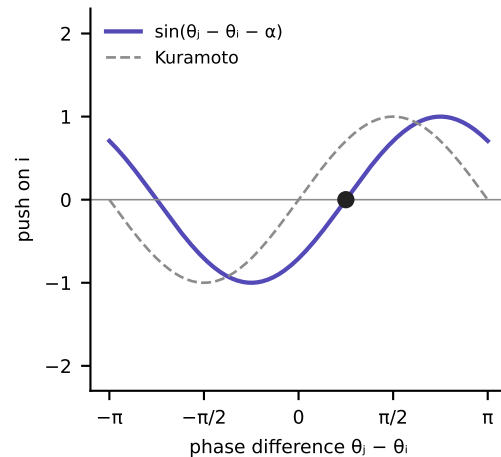
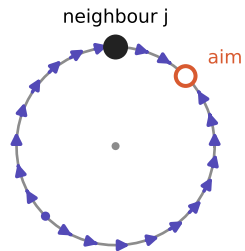
$K_{ij} \sin(\theta_j - \theta_i)$



each oscillator is pulled along the circle toward its neighbour's phase: at rest in phase, pushed away from opposite phase. The reference.

Kuramoto-Sakaguchi

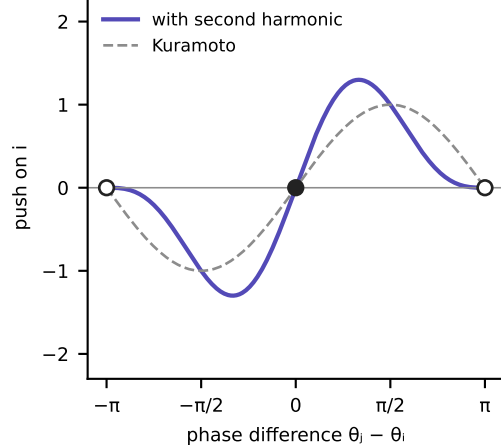
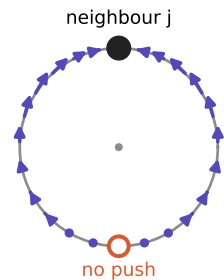
$K_{ij} \sin(\theta_j - \theta_i - \alpha), \alpha = \pi/4$



the pull aims α behind the neighbour: an in-phase pair still locks, but slowed, and a population can settle into travelling waves.

Second harmonic

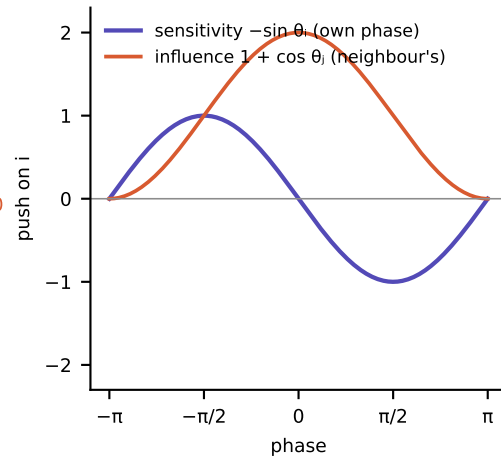
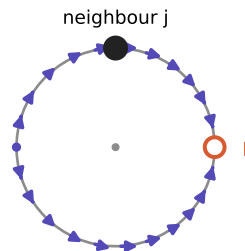
$K_{ij} [\sin(\theta_j - \theta_i) + \beta \sin 2(\theta_j - \theta_i)], \beta = 0.5$



adds a second harmonic: the pull toward the neighbour grows, and the push away from opposite phase vanishes, so pairs half a cycle apart are no longer driven apart.

Winfree

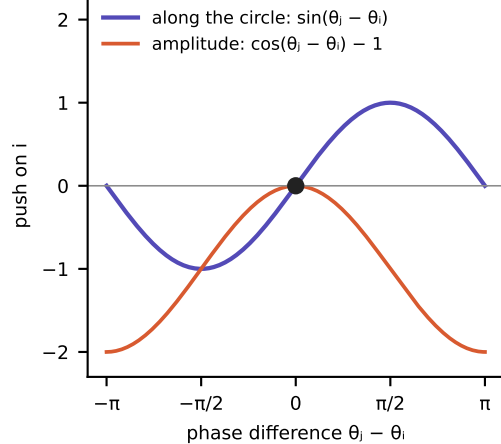
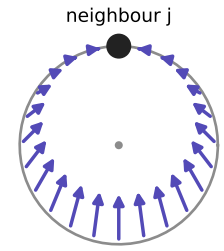
$-K_{ij} \sin \theta_i (1 + \cos \theta_i)$



every push points toward phase 0: the oscillator's own phase sets how much it responds, and the neighbour's phase only how strong the push is.

Stuart-Landau

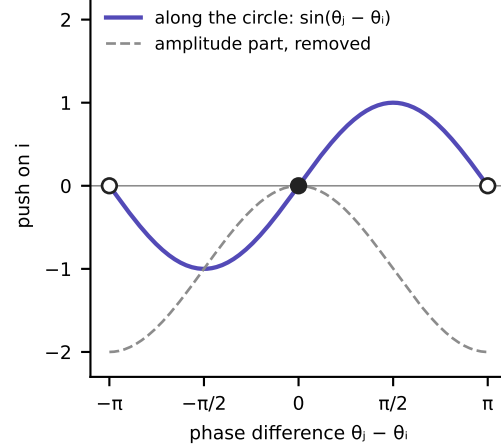
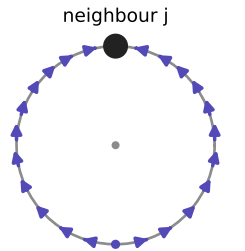
$K_{ij} (z_j - z_i), z = r e^{i\theta}, r$ the amplitude



each oscillator is pulled straight toward its neighbour in the plane: along the circle, as Kuramoto, and inward, so neighbours out of phase shrink each other's amplitude; amplitude then relaxes back to 1.

Stuart-Landau, fixed amplitude

the part of $K_{ij} (z_j - z_i)$ along the circle, $|z| = 1$



the amplitude is held at 1, so only the pull along the circle is left: Kuramoto again. It isolates what amplitude adds.

Left: the push an oscillator would feel at each phase around the circle from one neighbour j (black, at the top) with a positive kernel weight; arrows point the way it is pushed. Right: that push against the phase difference; filled dots are pairs at rest and stable, open dots at rest but unstable or neutral. A negative weight reverses every arrow, and each oscillator feels the sum over every other oscillator in its channel.